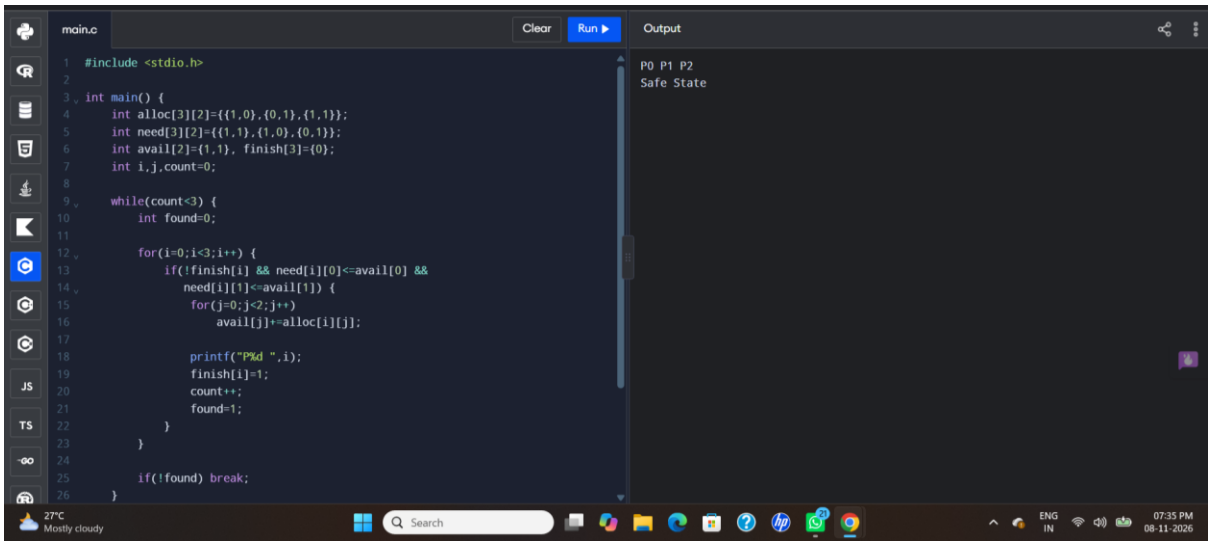
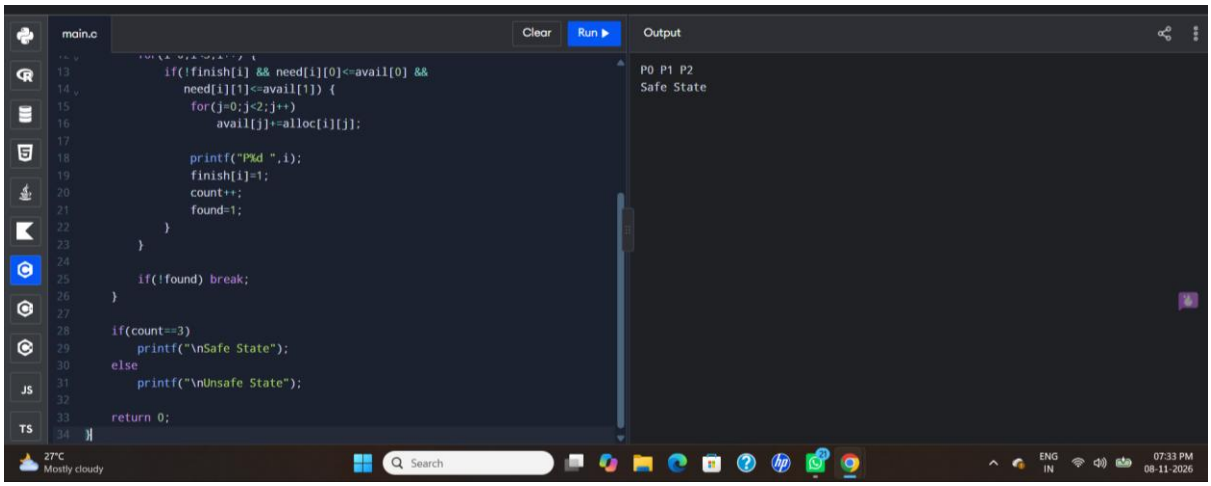
EXERCISE-17


main.c
Clear
Run
Output
#include <stdio.h>
int main() {
int alloc[3][2]={{1,0},{0,1},{1,1}};
int need[3][2]={{1,1},{1,0},{0,1}};
int avail[2]={1,1}, finish[3]={0};
int i,j,count=0;
while(count<3) {
int found=0;
for(i=0;i<3;i++) {
if(!finish[i] && need[i][0]<=avail[0] &&
need[i][1]<=avail[1]) {
for(j=0;j<2;j++)
avail[j]+=alloc[i][j];
printf("P%d ",i);
finish[i]=1;
count++;
found=1;
}
}
if(!found) break;
}
P0 P1 P2
Safe State
27°C
Mostly cloudy
Search
ENG
IN
07:35 PM
08-11-2026


main.c
Clear
Run
Output
if(!finish[i] && need[i][0]<=avail[0] &&
need[i][1]<=avail[1]) {
for(j=0;j<2;j++)
avail[j]+=alloc[i][j];
printf("P%d ",i);
finish[i]=1;
count++;
found=1;
}
}
if(!found) break;
}
if(count==3)
printf("\nSafe State");
else
printf("\nUnsafe State");
return 0;
}
P0 P1 P2
Safe State
27°C
Mostly cloudy
Search
ENG
IN
07:33 PM
08-11-2026